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Source-Learned Structural Priors for Chance-Constrained Simulation Optimization

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Abstract. We study chance-constrained simulation optimization when a policy is represented by an ordered profile and related source tasks are available before the target opens. A finite public function dictionary learns objective, constraint, and cumulative-risk structure from replicated source simulations. Paid target observations update the target function posteriors and the physical-risk family weights, while a correlated verification-aware knowledge-gradient rule allocates the remaining budget. On a declared development contract, observable trajectory exposures improve early mean prediction, and a paired source-structure control has higher mean final reward than a same-source full dictionary. The evidence is bounded: only three of eight development tasks improve, an independently frozen eight-task block does not replicate the mean ordering, a generic target-only RBF-KG control is at least as strong on both blocks, and learned risk has no independent deployment gain. We therefore claim a source-learned structural prior within the declared representation and dictionary as a development-block effect, rather than a stable deployment advantage, learned trajectory embeddings, factor-level risk recovery, or general high-dimensional optimization.

Key words: simulation optimization, transfer learning, structured priors, chance constraints, knowledge gradient
Subject classifications: Simulation: initial designs and independent verification; programming, stochastic-chance constrained
Area of review: Simulation

1. Introduction

Simulation optimization becomes difficult when a policy is written as a long ordered profile and each evaluation returns both a noisy objective and a probabilistic safety outcome. The profile may contain many coordinates, but the coordinates often arise from a smaller collection of response shapes. A new task therefore contains two kinds of uncertainty: which shapes matter in the target system, and how the target system changes their mean response and accumulated risk.

Historical simulations can guide both structural initialization and a source-to-target function prior. The project retains a frozen source-informed combined frontend that selects initial policies before target observations; its completed component evidence is reported in Section 5.3. A structural meta-prior learns which functions explain source responses, assigns a distribution to their coefficients and residual risk, and lets paid target observations update that distribution. These components address different decisions: the frontend chooses where to start, while the prior describes the response and risk functions updated during target search. This distinction matters in chance-constrained problems: source evidence may improve early prediction, yet a mismatched safety model can move the recommended policy in the wrong direction. The method must therefore preserve target correction, negative transfer, and independent verification in the same protocol.

We study this bounded problem in a declared public representation. Each policy is mapped to an eight-dimensional coordinate and evaluated through a finite dictionary of 68 centered and scaled functions. Separate source-task models describe the objective mean, the constraint mean, and the logarithm of cumulative risk variance. The target is evaluated with source-selected, same-source full-dictionary, and target-only controls. Target observations update the function posterior, while physical-risk family weights are updated from risk labels. A correlated Gaussian knowledge-gradient rule then values a candidate by its expected verified deployment utility under the remaining evaluation and verification budget.

The resulting information contract is deliberately narrower than a general representation-learning claim. The coordinate is declared rather than learned from trajectories; the dictionary is finite and not orthogonalized; and the risk head models total log variance rather than recovering separate physical common and local factors. These choices make the source-to-target path auditable and allow negative transfer to remain visible.

The paper makes three contributions. First, it specifies a reproducible source-to-target structural prior in which source observations learn function support and cross-task coefficient variation, while target observations can remove a mismatched structure. Second, it couples this posterior to a verification-aware decision rule and records the distinct roles of prediction, candidate ranking, and independent certification. Third, it evaluates the contract under paired source archives, initial designs, target budgets, and verification streams. The results show an early-budget advantage under matched mechanisms, a small and unstable contribution from learned risk structure, and no universal advantage for the knowledge-gradient or for operational deployment.

This separation gives a precise interpretation of the evidence. A reduction in prediction error is not treated as a deployment certificate; a small knowledge-gradient increment is not treated as proof of a general sampling advantage; and an infeasible target task is retained as a failure of the end-to-end deployment protocol. The conclusions therefore apply to the declared representation and finite dictionary, while trajectory encoders, factor-level risk recovery, and general high-dimensional optimization remain open problems.

2. Related Literature

Simulation optimization studies expensive stochastic systems whose objective or constraints are available only through simulation (Fu 2002, Amaran et al. 2016). Stochastic kriging separates response uncertainty from simulation noise (Ankenman et al. 2010), while constrained Bayesian optimization models noisy feasibility observations jointly (Letham et al. 2019). We study the additional transfer question: how should a finite source archive influence the target posterior before the target has many observations?

Transfer Bayesian optimization includes learned initializations (Wistuba et al. 2015, Feurer et al. 2022), hierarchical and deep-kernel priors (Wistuba and Grabocka 2021, Rothfuss et al. 2021a,b), and multi-task hyperparameter models such as HyperBO (Wang et al. 2024). Our transfer object is narrower. We learn a finite function prior from replicated source responses, then update that prior with paid target observations. The

method does not retrieve related tasks from an unrestricted repository and does not reuse a source optimum as a target label.

Chance-constrained ranking and selection separates search from a final independent decision (Hong et al. 2015, Boesel et al. 2003). We retain that separation: the posterior ranks candidates, while a fresh verifier makes the deployment decision. The resulting claim is conditional on a declared ordered-profile representation and source-target task law; no distribution-free optimizer is claimed (Wolpert and Macready 1997).

3. Source-Learned Structural Prior

3.1. Ordered policies and observable coordinates

Let $h : [0,1] \to [0,1]$ be an ordered policy profile. A public candidate x is represented by an eight-dimensional coordinate $\psi(x)$ constructed from the declared trajectory and cost exposures. This coordinate is observable before target outcomes are collected. It is not an encoder trained from unobserved target trajectories. A nominally long profile is therefore treated as a discretized ordered rule, rather than as an arbitrary vector of unrelated free variables.

For target task q, one replication returns an objective $Y_q^f(x)$ and a constraint observation $Y_q^g(x)$. We minimize $f_q(x) = \mathbb{E}Y_q^f(x)$ under the chance constraint

$$\Pr\{Y_q^g(x) \le 0\} \ge 1 - \alpha. \tag{1}$$

The environment's cumulative risk is represented by $\eta_q(x) = \log \mathrm{Var}(Y_q^g(x))$. The risk head describes total log variance; it does not claim to identify separate physical common and local shocks.

3.2. Finite function model

Each source head $t \in \{f, g, \eta\}$ uses the declared finite dictionary $\{\phi_k(\psi)\}_{k=1}^{68}$:

$$H_{q,t}(x) = b_{q,t} + \sum_{k=1}^{68} \beta_{q,t,k}\phi_k(\psi(x)) + r_{q,t}(x). \tag{2}$$

The dictionary is centered and scaled using only the public candidate bank. Source tasks estimate which functions are useful and the cross-task coefficient variation. The target coefficient mean for non-intercept functions is zero; source information enters through the covariance and the structural support. The residual $r_{q,t}$ is epistemic model error and is kept separate from environmental risk.

The target arms differ in the declared source information: a source-selected support, the same-source full dictionary, or a target-only full dictionary. A paid target observation y updates the Gaussian function posterior,

$$\theta_j^+ = \theta_j^- + K_j(y - \hat{y}_j), \tag{3}$$

where the observation noise is recorded separately for each head. The physical-risk head additionally maintains a finite factor-family particle mixture; its weights are updated by the exact Gaussian log-sample-variance likelihood. This separates mean-structure correction from risk-family reweighting and leaves negative transfer visible in the paired controls.

3.3. What the prior does not claim

The coordinate is declared, the dictionary is finite and non-orthogonal, and the risk head models total variance. Consequently, the present method does not establish learned trajectory representations, separate factor recovery, stable deployment superiority, or optimization over general independent high-dimensional variables. Those are distinct research questions rather than implicit consequences of (2).

4. Verification-Aware Decision Rule

After each paid target batch, the posterior supplies a predictive mean and covariance for (f, g, η) over the public candidate bank. A candidate's deployment value is the expected objective quality multiplied by the probability that an independent verifier will certify its constraint. For a fixed recommendation x, the verifier uses fresh replications and the frozen chance level α; search observations never enter this test.

The acquisition rule evaluates the expected increase in this verified utility under the remaining target and verification budget. It uses the correlated Gaussian approximation for the joint mean–risk posterior and finite quadrature for the fantasy observation. The live physical-risk posterior updates factor-family weights; the mean function posteriors are updated by Gaussian conditioning. The fantasy calculation uses the frozen approximation analyzed in the A45 numerical diagnostic. That diagnostic found a maximum 9-versus-13-node difference of 2.93×10^{-5} in the controlled risk-only probe, so the existing acquisition rule is retained for the reported pilots.

This utility is a decision rule, not a theorem that the selected policy is the global optimum of the chance-constrained problem. We report the initial recommendation, final recommendation, true post-decision reward, independent certificate, and false-certificate indicator separately. A successful certificate requires a fresh verifier; a posterior feasibility probability is never reported as a deployment certificate.

5. Experimental Contracts

5.1. Source archive and target protocol

The frozen M1 archive contains eight source tasks, 64 public policies per task, and 12 independent repetitions per policy (6,144 source episodes). The target uses the same public candidate bank, eight common initial policies, 24 search points, and 928 independent verification replications. All arms share initial points, target seeds, keyed observation streams, and the verification procedure.

The primary comparisons separate four questions: (i) prediction after the common initial design, (ii) target correction of a mismatched source structure, (iii) risk calibration, and (iv) final verified decision utility. Source, target, search, and verification calls are accounted for separately.

5.2. Feasible-deployment development contract

The original Inventory law often produced no feasible policy in the 96-point bank. A separate development contract therefore fixes the public constraint price range to $[0.4, 0.9]$, while retaining the demand schedule, capacity, budget, candidate bank, and verifier. Its eight development law seeds have 19–75 feasible candidates and feasible-objective spans of 0.053–0.130 under the oracle screen. This screen is a task-contract preflight; it is not available to the learner. The original confirmation interval 95433–95440 was not run as a block, but seed 95434 overlaps the development artifact and is therefore excluded from any clean-confirmation claim. A separate v2 contract freezes eight new law seeds (95800–95807), records an oracle-only preflight before target observations, retains every seed, and reports the resulting independent block without reading the original block.

The final attribution control compares source-selected structure with the same-source full dictionary under identical observable coordinates and pooled risk. It uses the same eight tasks and the same keyed target stream. This is development evidence, not evidence that the source-structure effect generalizes to an independent block.

5.3. Retained structural-initialization evidence

The frozen combined legacy frontend has a completed component study on FactorShock, InventorySupplyChain, and QueueResourceControl. At nominal dimension 1,000, each family contributes 20 algorithmic seeds. The frontend uses 384 reusable source calls per held-out-family archive to freeze ten target proposals. Proposal-only selection uses ten target search calls; Stacked GP and canonical SAASBO add three adaptive calls. All three frontend arms achieved 60/60 true-feasible and independently certified recommendations with zero false certificates. Replacing the frontend by common Sobol gave 0/60, 1/60, and 0/60 certifications, respectively (Table 1). This frontend combines universal policy shapes with source templates through the historical `LearnedMetaPrior` implementation; it is distinct from the later public 64-profile `SourceScoredProfileAtlas`. Component ablations certified 40/60 with source templates alone (0/20 FactorShock, 20/20 Inventory, 20/20 Queue), and 23/60 with universal shapes alone (20/20, 2/20, 1/20). The combined result therefore reflects complementary components rather than a contribution attributable wholly to source learning.

The native transfer matrix retains each comparator's own initialization and adaptation with the 384-call source archive and 13-call target search budget. Its best certification totals were RGPE's 30/60 and MetaBO's 28/60. In the separate common-frontend backend comparison, the eight transfer methods reached 56–60/60. These comparisons retain both the native and supplied-frontend pipelines; the common-Sobol factorial above supplies the direct frontend intervention for the two fitted backends. The supplement preserves the individual method totals and evidence locations.

Table 1 Historical frontend component study. Source calls are per reusable archive and search calls are per target run; terminal verification calls are additional and separately recorded. These results use the common historical v69 Gaussian terminal protocol, not the current physical-Inventory verifier.

Frontend	Backend	Source	Search	Certified	False
Combined legacy frontend	Proposal only	384	10	60/60	0
Combined legacy frontend	Stacked GP	384	13	60/60	0
Combined legacy frontend	Canonical SAASBO	384	13	60/60	0
Common Sobol	Proposal only	0	10	0/60	0
Common Sobol	Stacked GP	384	13	1/60	0
Common Sobol	Canonical SAASBO	0	13	0/60	0

This study supports the value of structural initialization on these three fixed task families. Its seeds measure algorithmic repeatability, not generalization across 60 independent task laws. The current physical Inventory model has four state-bin actions and replicated mean/risk labels; it uses a common eight-policy initialization rather than the combined frontend. Consequently, the frontend remains a demonstrated component of the project, while independent decision superiority of the combined frontend–function-prior–KG pipeline remains unestablished. A development integration check reuses the historical frozen proposals and source archives at seed 80 in each family, with ten common single-episode observations and three subsequent raw-observation KG calls. The source-orthogonal and same-source full priors do not improve the deployed objective over proposal-only selection. Both can recommend infeasible unobserved Inventory candidates after ten calls; a common independent verifier retaining initial-policy support preserves the deployed result. Under the declared primary-certificate utility and common finite candidate bank, the retained initial deployment leaves no utility headroom in FactorShock and only 0.0021 and 0.0014 in Inventory and Queue, respectively. These post-decision bounds apply to this candidate bank and proxy utility. This check uses a declared primary-certificate utility for acquisition and a guarded shortlist for deployment; it diagnoses the remaining mismatch between these objectives, rather than establishing a unified deployment KG advantage. Its replay entry point is given in the supplement.

5.4. Negative and external controls

We retain the original adverse Inventory law, the physical-risk factor-family diagnostics, the official HyperBO interface and its no-feasible-target loop, and the legacy atlas/Energy studies as separate evidence identities. These controls are not pooled into a single performance table because their source archives, initial designs, risk heads, or verification contracts differ.

Table 2 Paired source-structure attribution on the eight-task development block.

Arm	Tasks	Mean final reward	Certified	False certificates
Source-selected structure	8	0.86127	8/8	0
Same-source full dictionary	8	0.85582	8/8	0

6. Results

6.1. Early prediction and target correction

On the feasible-deployment development block, observable exposure reduced the eight-point holdout RMSE for the objective mean from 0.594 to 0.270 and for the constraint mean from 0.204 to 0.141. Constraint-ranking Spearman correlation rose from 0.119 to 0.624. This is a prediction result; the initial true recommendation was feasible in 7/8 tasks for both representations and their mean initial rewards were essentially identical.

The risk factor-family prior also improved total log-risk prediction. In the source leave-one-task-out check its mean log predictive density gain over a pooled-risk control was 0.692, and all eight development targets had lower initial η RMSE (0.301 versus 0.363). The coefficient RMSE, however, was not lower (0.255 versus 0.251), so the evidence supports total-risk calibration rather than separate physical-factor recovery.

6.2. Decision attribution

The observable representation versus fixed policy-coordinate closed loop gave mean final true rewards 0.8613 and 0.8538, respectively. Only 3/8 tasks had a positive search increment, so the predeclared stable-deployment gate failed. Adding the learned risk prior to observable exposure reduced mean final reward slightly to 0.8608; one task declined and the others were unchanged.

The new same-source control isolates the mean structural prior on the development block:

The paired final-reward differences were positive on 3 tasks, negative on 1, and zero on 4, with mean +0.00545. Thus source structure has a measurable mean-prior contribution in this block, but not a stable deployment advantage.

6.3. Independent fixed-seed confirmation

The separate v2 block retained all eight predeclared law seeds after its oracle-only preflight. The source-selected arm had mean final true reward 0.44274 (5/8 certified), while the same-source full dictionary had 0.45160 (4/8 certified); neither arm produced a false certificate. The paired differences were positive on 2 tasks, negative on 2, and tied on 4. Thus the development-block ordering is not confirmed by this independent block. The original 95433–95440 interval was not run as a block; seed 95434 overlaps the development artifact and is excluded from any clean-confirmation claim. The v2 block is a separate contract and does not repair that overlap retroactively.

All eight fixed seeds were retained after the oracle-only preflight; paired final-reward differences were positive on 2, negative on 2, and tied on 4 tasks.

Table 3 Independent fixed-seed confirmation block; all eight tasks retained.

Arm	Tasks	Mean final reward	Certified	False certificates
Source-selected structure	8	0.44274	5/8	0
Same-source full dictionary	8	0.45160	4/8	0

Table 4 Same-contract target-only baseline and paired mean differences.

Block	Source-selected mean (certified)	Full dictionary mean (certified)	Target-only mean (certified)	RBF-KG mean (certified)	Source–RBF mean difference
Development	0.86127 (8/8)	0.85582 (8/8)	0.76199 (7/8)	0.86239 (8/8)	-0.00112
Independent v2	0.44274 (5/8)	0.45160 (4/8)	0.45148 (4/8)	0.45912 (4/8)	-0.01638

6.4. Same-contract target-only baseline

The target-only full-dictionary control uses the same 68-function dictionary, candidate bank, initial design, paid-observation budget, keyed noise streams, pooled factor-family risk arm, acquisition rule, and independent verifier; it starts without source labels. Its mean final true rewards are 0.76199 on the development block and 0.45148 on v2, compared with 0.86127 and 0.44274 for source-selected structure. The corresponding same-source full-dictionary means are 0.85582 and 0.45160. The source-selected minus target-only difference is therefore +0.09928 in development but -0.00873 in v2. This feasible nonempty baseline completes the information-contract comparison and does not restore an independent deployment claim.

We also ran a generic target-only RBF-KG baseline with fixed isotropic lengthscale 0.25 and outputscale 1.0, using the same observable coordinates, budget, keyed streams, and verifier but no source labels or source-risk prior. Its means are 0.86239 on development and 0.45912 on v2, exceeding the source-selected arm by -0.00112 and -0.01638, respectively. Thus the source structure is attributable relative to the deliberately full dictionary in the development block, but it is not superior to a generic target-only RBF control.

6.5. Independent verification and negative evidence

All 24 cells of the observable/risk development loop and all 16 cells of the source-structure control received valid independent certification with no false certificates. This verifies the task and verifier contract; it does not turn the development block into confirmation. The original physical-risk closed loop had seven of eight targets with no feasible candidate in the fixed bank, and the official HyperBO comparison used a target with no feasible candidate. Those failures remain part of the result because candidate coverage and task feasibility are part of an end-to-end deployment claim.

The risk-only A45 diagnostic found that refreshed mixture weights, moment collapse, and quadrature-node choices changed probe rankings only modestly; it did not identify the current Gaussian fantasy approximation as the cause of the weak deployment increment.

7. Discussion and Limitations

The evidence supports a bounded statement. In a declared ordered-profile representation and finite dictionary, replicated source tasks can learn a useful response-function structure. A small common target design can then improve early mean prediction, and target evidence can reduce the weight of a mismatched structure. The paired full-dictionary control shows that this effect is attributable to source-selected function structure on the checked development block.

The evidence does not support three stronger statements. First, observable exposures are not a learned trajectory representation; earlier encoder and state-coordinate gates did not produce a transferable end-to-end gain. Second, improved total log-risk prediction did not yield a risk-specific deployment gain, and coefficient recovery was not improved. Third, the independent deployment gate was not passed: only 3/8 source-structure tasks improved, while the learned-risk arm had a small negative mean difference. The generic target-only RBF-KG control also exceeded the source-selected arm by 0.00112 on development and 0.01638 on v2, so the source structure is not a generic constrained-BO improvement under this contract.

The task law is synthetic and the candidate bank can miss the safe region. The feasible-deployment range was declared before the development block. The original 95433–95440 confirmation interval was not run as a block, but seed 95434 overlaps the development artifact and is not eligible as untouched confirmation. The separate v2 block retained all eight new seeds but did not reproduce the development ordering. The source archive is assumed to be available; retrieval cost and operational value are not solved. The finite dictionary is not an arbitrary high-dimensional optimizer, and the legacy atlas and Energy results answer a different initial-design question. Finally, zero false certificates in these cells is evidence about the declared verifier and replication law, not a universal safety guarantee.

8. Conclusion

We presented a source-learned structural prior for chance-constrained simulation optimization of ordered policy profiles. The method transfers a finite function structure, updates it with paid target observations, and separates search utility from independent deployment verification. The current evidence supports early prediction and a bounded attribution effect under matched mechanisms, with negative transfer and task infeasibility retained. The generic target-only RBF-KG control is at least as strong on both reported blocks, so no generic decision-superiority claim is made.

The paired development control shows a contribution from source-selected mean structure relative to a same-source full dictionary, but the separate fixed seed block does not confirm its mean ordering. The result is therefore not a stable independent-deployment claim. Learned trajectory coordinates, factor-level risk recovery, and general free-variable high-dimensional optimization remain outside the evidence. A submission should state the structural-prior contribution at this declared scope and report the unresolved deployment and representation limits explicitly.

9. Data and Code Accessibility

This study uses no proprietary or human-subject data. The synthetic inventory generator, public candidate bank, source archive, frozen contracts, target-only baseline results, paired bootstrap summaries, manuscript table generators, and the two compiled source files are included in the submission package under `submission/meta_prior_or_package_v1/`. The package README gives the fixed-entrypoint replay command `python3 performance/replay_meta_prior_submission.py -compile` and records the artifact paths used by every reported table. The original confirmation interval is intentionally not read by that replay; its overlap with development at seed 95434 is recorded in the manifest.

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